AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 28 · STUDENT LESSON

Prompt Experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 12: use the same scoring rules for comparisons.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of controlled prompt experiments. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of controlled **prompt** experiments.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

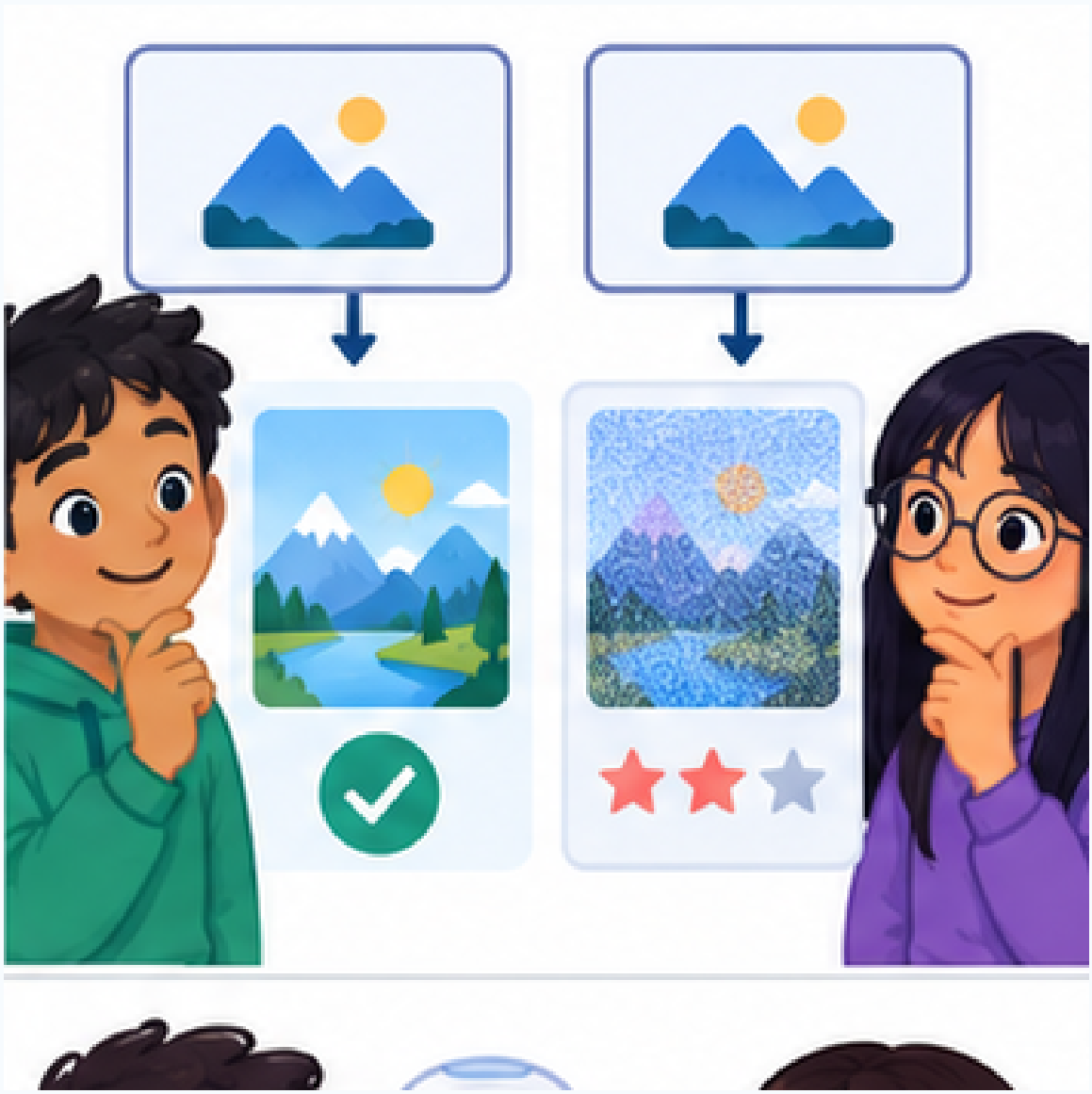
Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 28: **Prompt** Experiments

Prompt quality is evaluated by **controlled comparison**s and success criteria.



THINKER MISSION

Change one **prompt** variable at a time and score results.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Structured **output**s and validation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can compare **prompt**s on **paired case**s and inspect a subgroup before choosing.

STUDENT LESSON · 2 OF 18

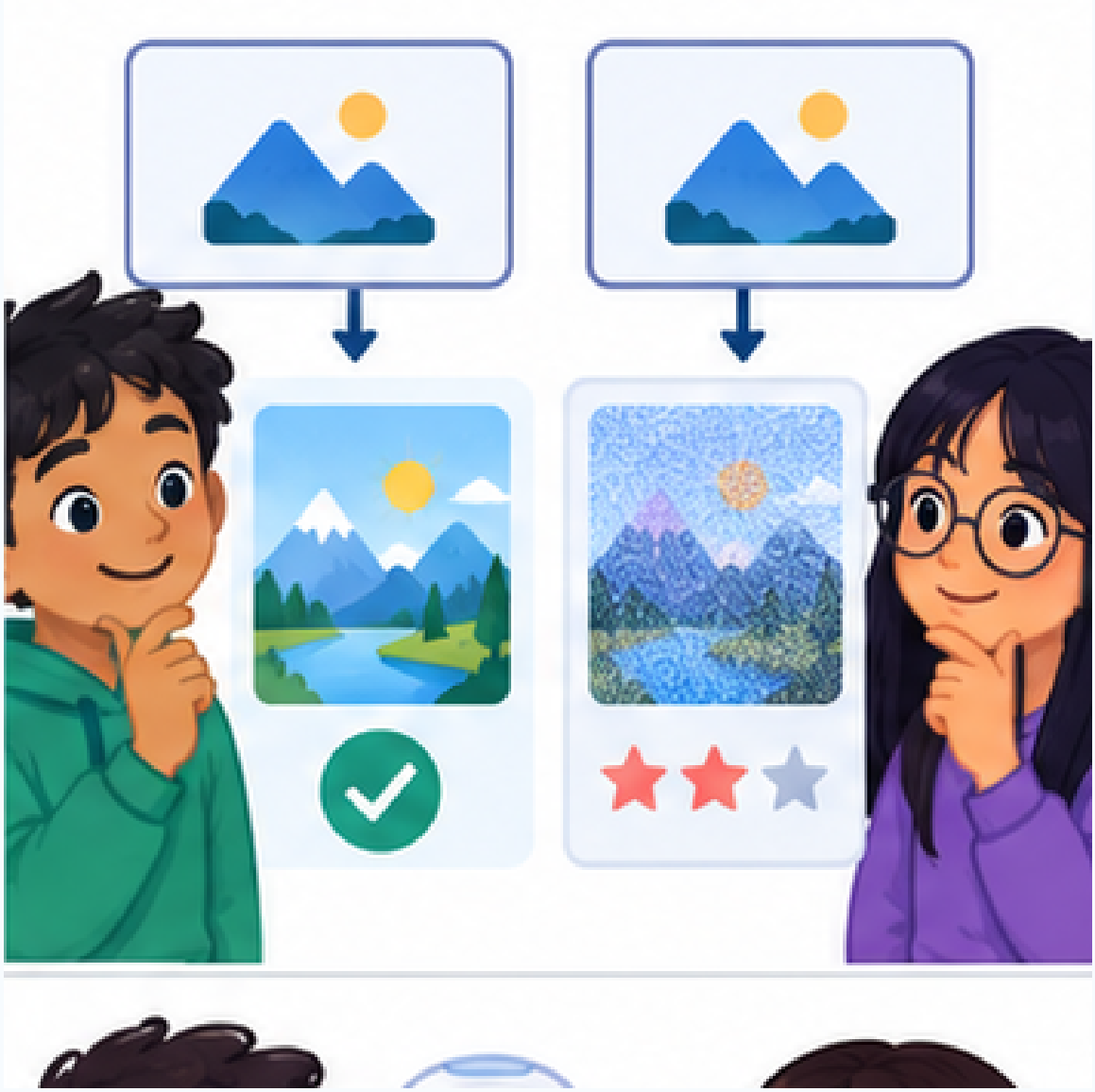
Notice the evidence

> Investigate the supplied Prompt Experiments case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **prompt** experiments to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A **prompt** experiment compares complete specifications using the same task cases. Define success before inspecting **output**s; change one intended instruction and preserve other settings.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

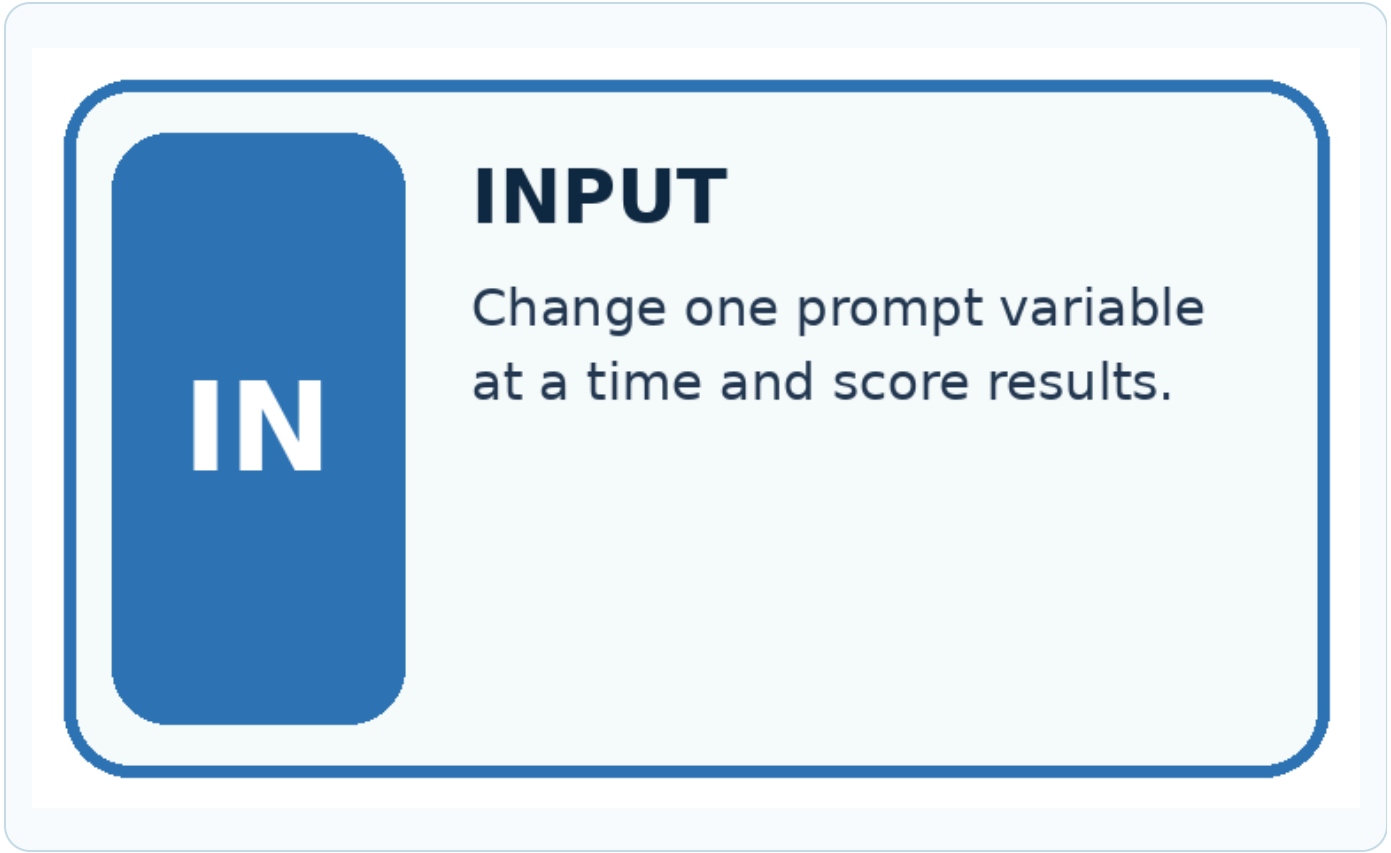
See how it works

> Follow the mechanism for Prompt Experiments. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

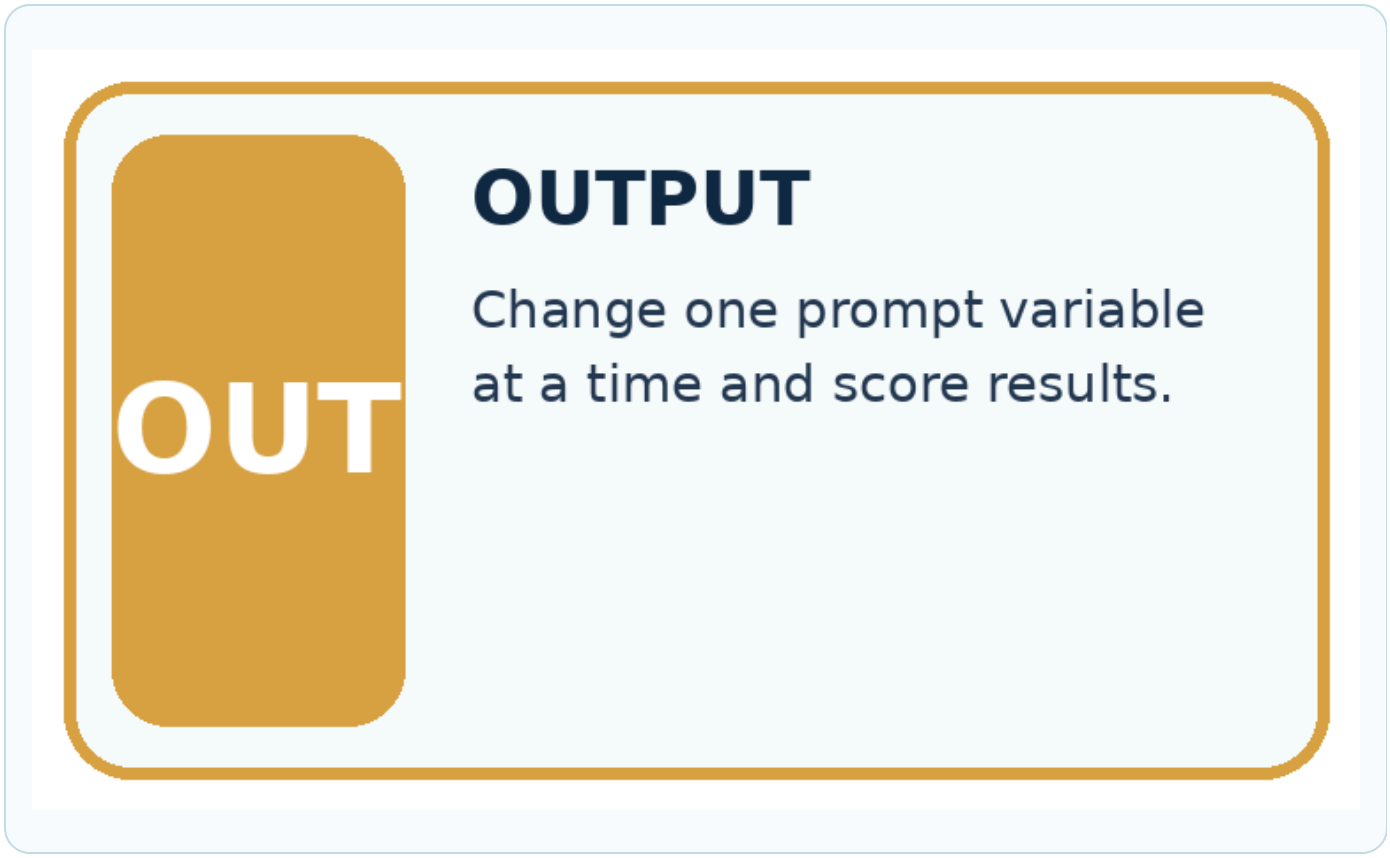
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Prompt quality is evaluated by controlled comparisons and success criteria.

↓


OUT
OUTPUT
Change one prompt variable at a time and score results.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Evaluate both **prompt**s on **paired case**s with fixed model/version, source records, **output** contract and scoring **rubric**. Use development results for selection, then reserve new final cases.

Score each case rather than counting attractive sentences.

Controlled prompt experiments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.	Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric.	P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

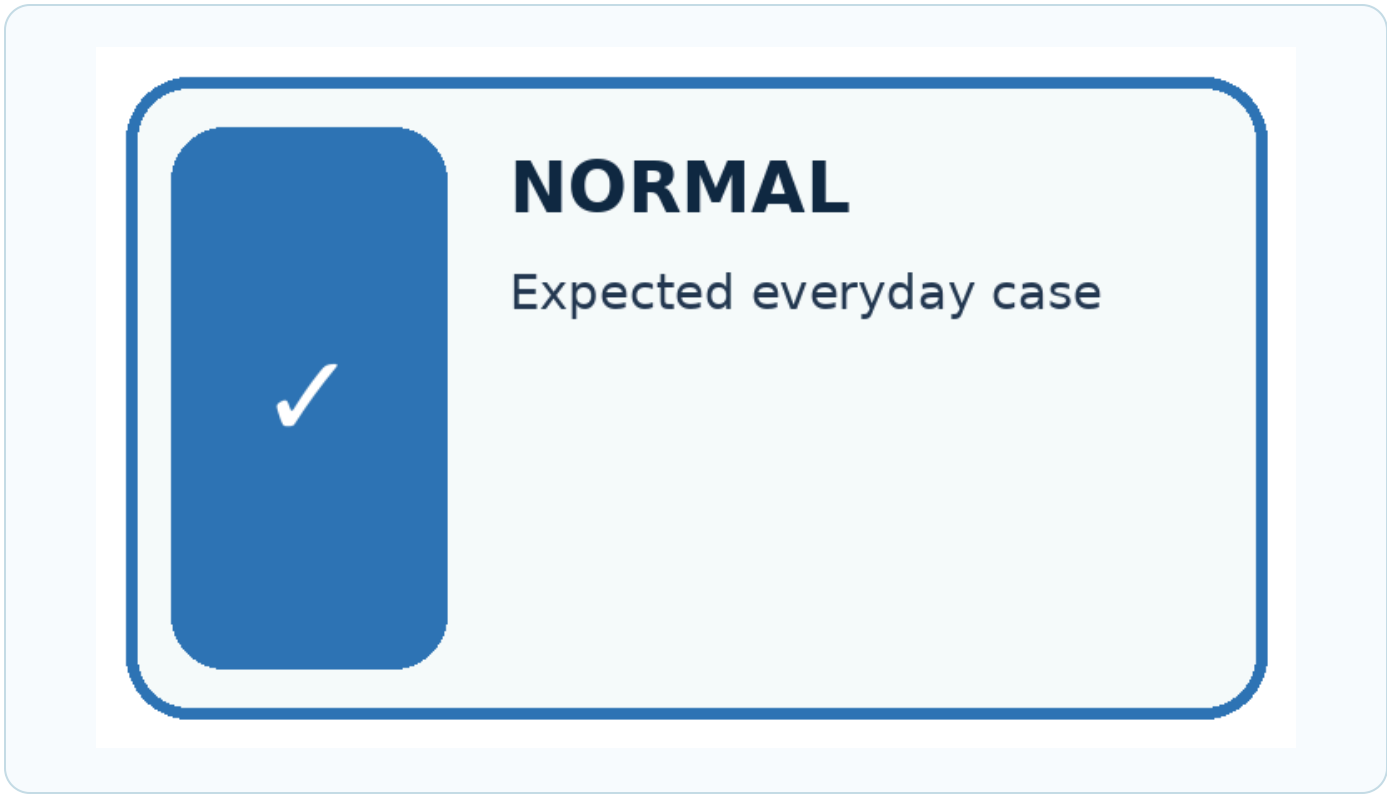
> Compare the supplied cases for Prompt Experiments. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

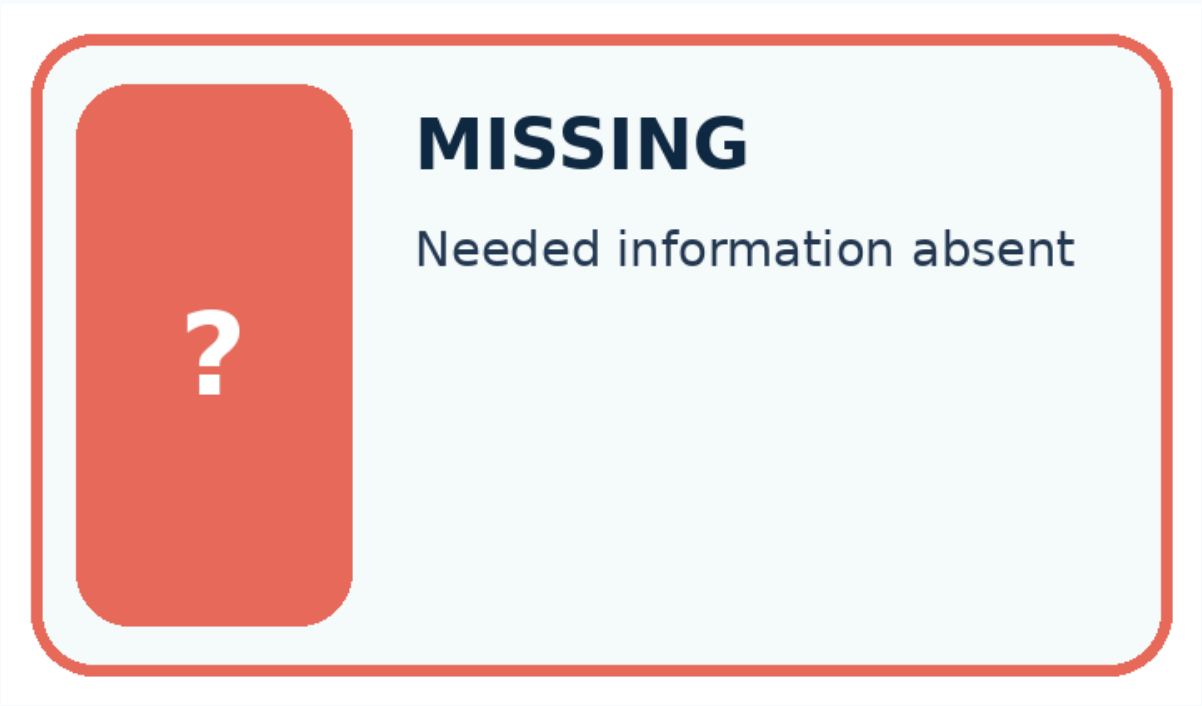
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Make missing-source correctness a required selection gate.

Predicted result

Prediction: Neither **prompt** clears a 2/2 gate on these records.

Why the result follows

Reason: P fails 7 and Q fails both 4 and 7; both need revision.

STUDENT LESSON · 5 OF 18

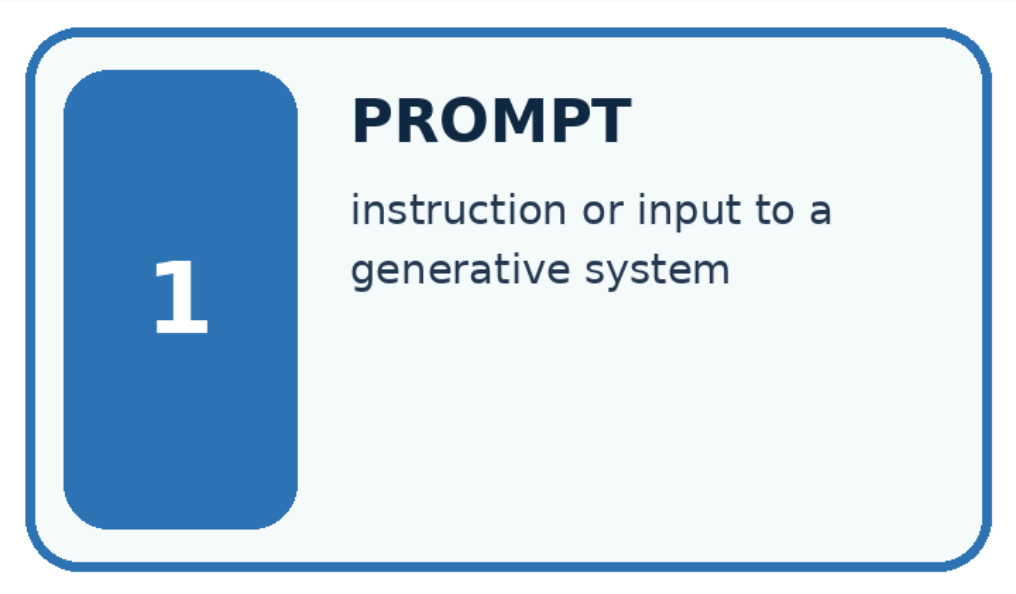
Words that unlock the idea

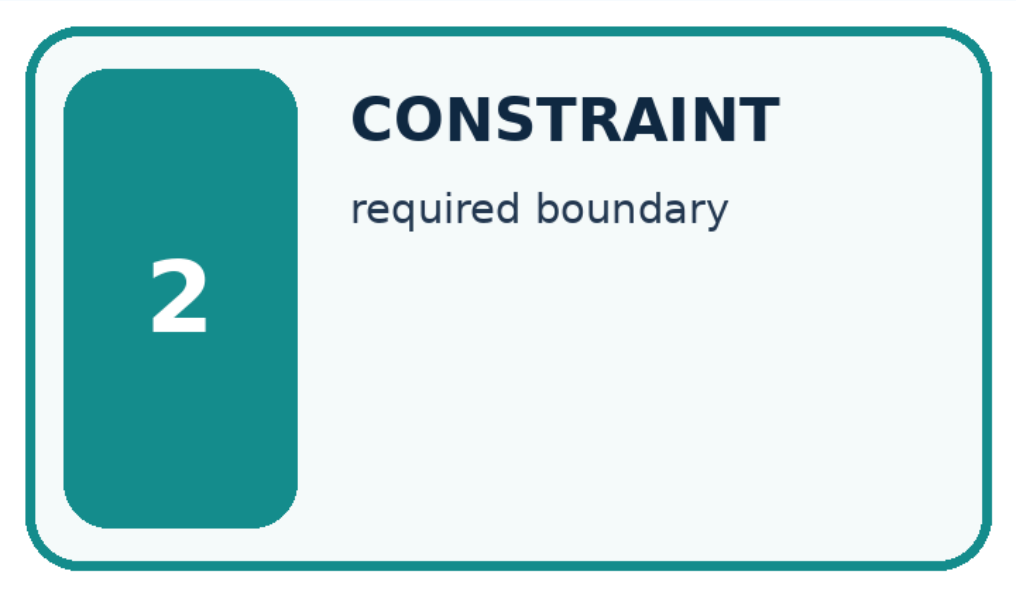
> Use the vocabulary of Prompt Experiments to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
PROMPT
instruction or input to a generative system


2
CONSTRAINT
required boundary





Word	My own example	Not this / boundary
prompt	Input instructions and context supplied to a generative system.	
constraint	A condition that a valid solution must satisfy.	
criterion	One stated standard used to judge a result.	

Word	My own example	Not this / boundary
controlled comparison	A comparison designed to keep relevant conditions alike except for the factor being studied.	

Words used in the closer explanation

Word	Meaning
Paired case	The same case evaluated under each compared procedure
Rubric	Prespecified criteria for judging an output
Regression	A previously passing condition that fails after a change

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Prompt Experiments, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 28 • WORKED EVIDENCE

1. Prompt A meets 3 of 6 rubric checks.
2. Add one constraint: ‘Use 3 sentences and name no ending.’
3. Prompt B meets 5 of 6 checks.
4. Improvement = +2 checks; repeat on more books before claiming it generalizes.

1. **Prompt** A meets 3 of 6 **rubric** checks.

2. Add one **constraint**: ‘Use 3 sentences and name no ending.’

3. **Prompt** B meets 5 of 6 checks.

4. Improvement = +2 checks; repeat on more books before claiming it generalizes.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results.

P passes 4/6 and Q 5/6. P passes one of two missing-approval cases, Q also one of two. Q improves the aggregate while leaving this condition unchanged.

Worked explanation · Prompt Experiments

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prompt Experiments

A fictional book-help project tests two **prompt** versions on seven shared development cases.

The facts we will use

Supplied facts

P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Supplied facts

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

AI ACADEMY / AI-2/28

Paired overall and missing-source passes

	Overall passes /7	Missing-source passes /2
P	4	1
Q	5	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/28

Prompt Experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired setup

Use the same seven cases and pass criteria for both prompts.



2. Overall rates

P=4/7; Q=5/7.



3. Missing-source slice

P passes 1/2; Q passes 0/2 in cases 4 and 7.



4. Decide explicitly

Q improves overall count but worsens missing-source behavior; use the declared priorities.

CHANGE ONE THING

Make missing-source correctness a required selection gate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Paired setup	Use the same seven cases and pass criteria for both prompts.
2. Overall rates	P=4/7; Q=5/7.
3. Missing-source slice	P passes 1/2; Q passes 0/2 in cases 4 and 7.
4. Decide explicitly	Q improves overall count but worsens missing-source behavior; use the declared priorities.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

What this example does not establish

Keep the stated limit

Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

Keep the stated limit

Read the labelled picture: Name the case P passes and Q fails, and explain why it matters.

Keep the stated limit

Change one thing: Make missing-source correctness a required selection gate. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Prompt Experiments claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: One impressive output proves a prompt is reliable. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

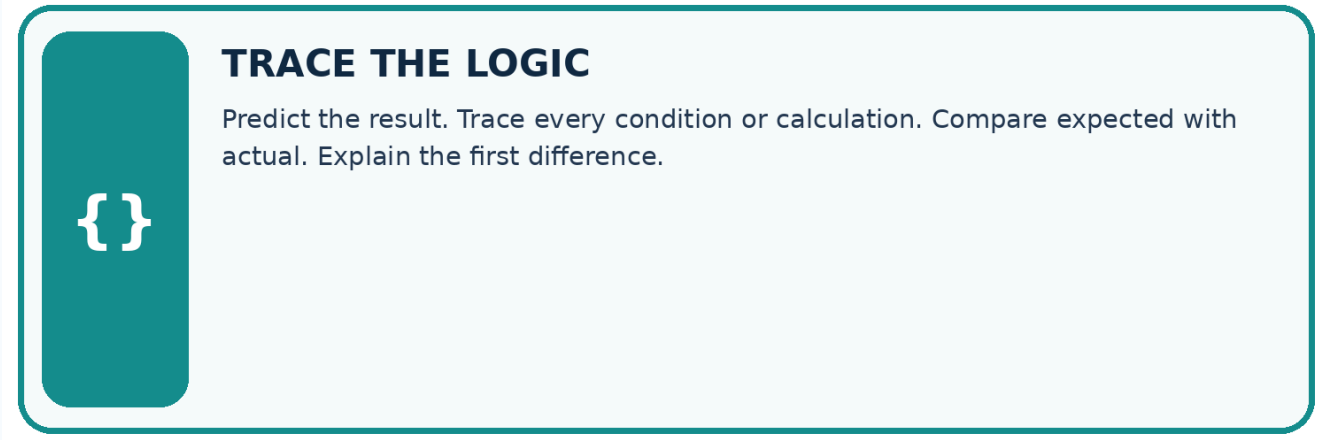
Practise together

Practise Prompt Experiments with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Prompt selection on development cases cannot be reported as independent final performance.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Controlled **prompt** experiments”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Controlled **prompt** experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Name the case P passes and Q fails, and explain why it matters.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

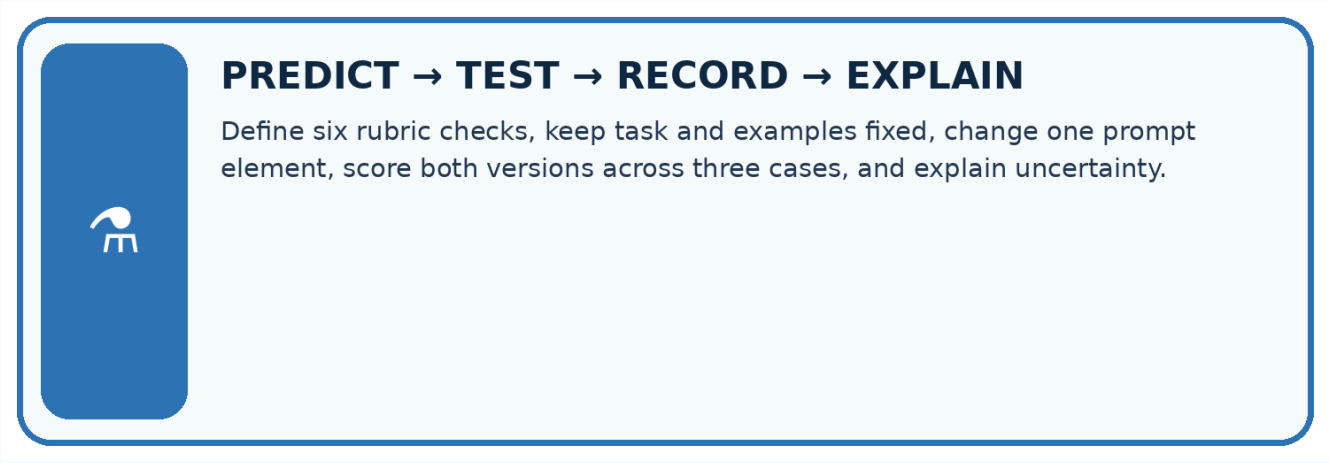
Run the investigation

> Investigate Prompt Experiments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Define six **rubric** checks, keep task and examples fixed, change one **prompt** element, score both versions across three cases, and explain uncertainty.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

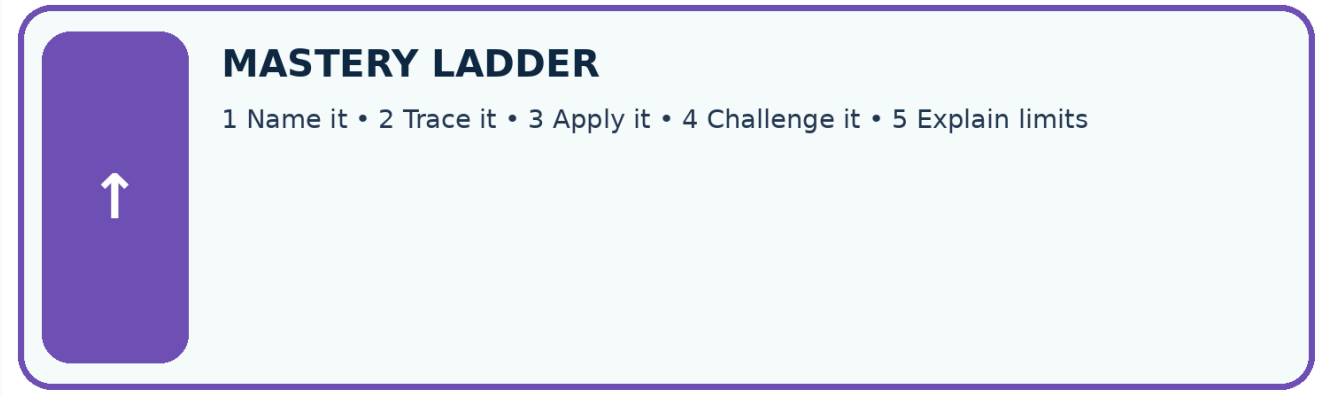
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Prompt Experiments. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

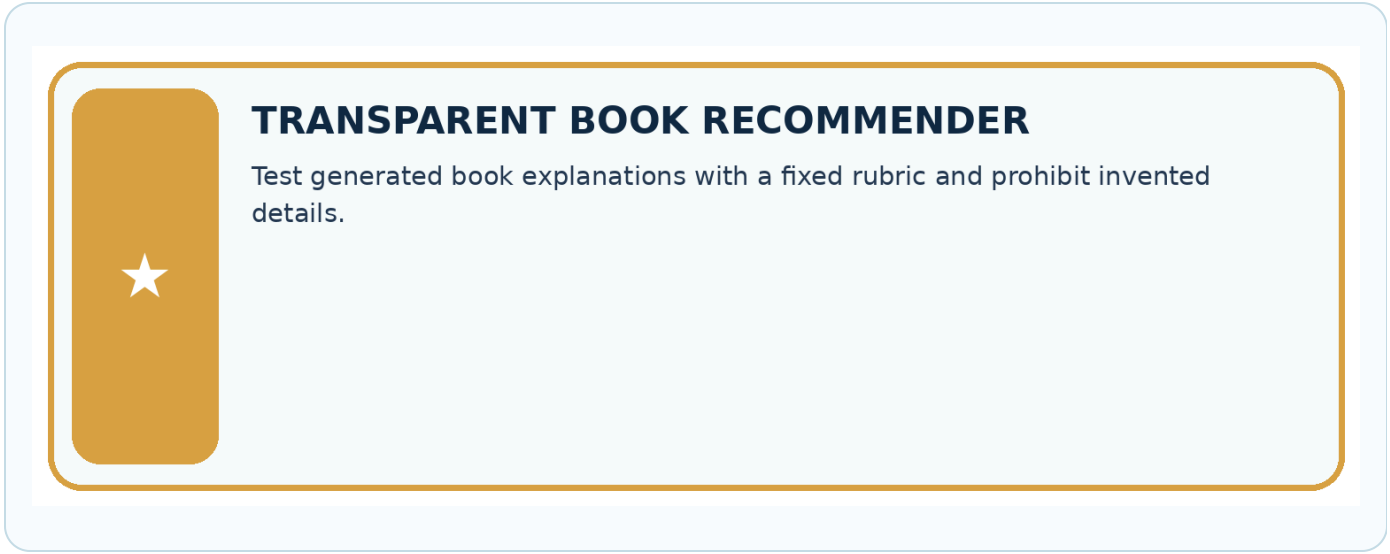
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Test generated book explanations with a fixed **rubric** and prohibit invented details.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately.

STUDENT LESSON · 12 OF 18

Show what you know

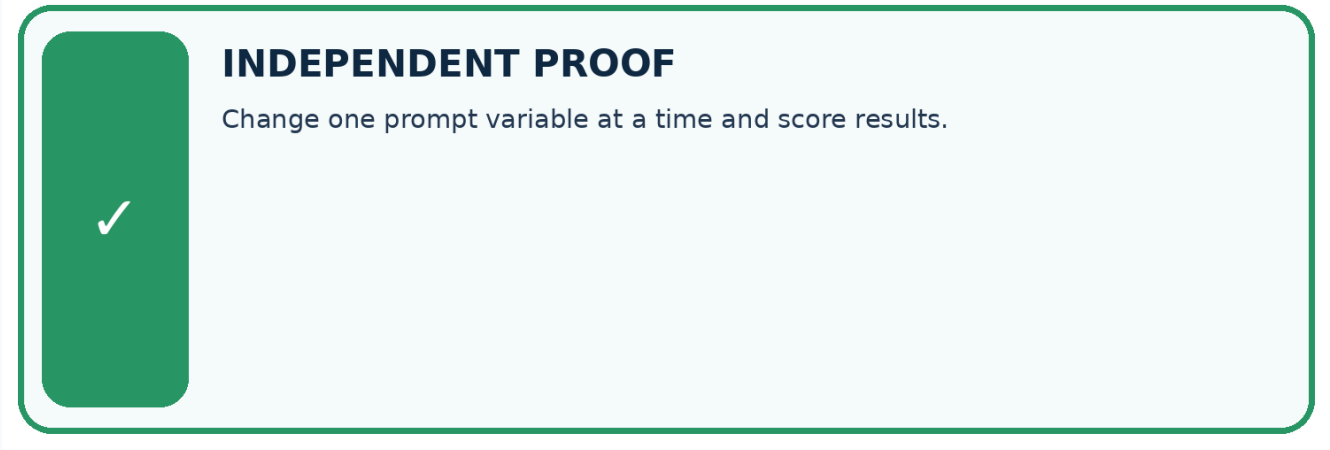
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Change one **prompt** variable at a time and score results.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Prompt Experiments case for supported practice. Repeated examples do not become fresh assessments.

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

The question

Design a fair comparison.

Worked explanation

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare prompts under the same conditions

Evidence or stage	Value or result
Prompt A	Original catalogue request
Prompt B	Add under 100 pages and explain the match
Hold fixed	Catalogue, requests and scoring
Check	Constraint compliance and supported explanation

Prompt B adds under 100 pages and explain the match to **Prompt** A. Both use a catalogue.

Design a fair comparison.

Use identical requests, catalogue and scoring criteria. Check **constraint** compliance and supported explanation; repeat stochastic **output**s when relevant.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Prompt A and B are tested on different questions. B scores higher. Can wording alone explain the difference? Design a **controlled comparison**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Test generated book explanations with a fixed **rubric** and prohibit invented details.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Prompt A is tested on easy requests and B on requests with missing data. B scores lower. Design a fair comparison and specify what to do when **output**s vary across runs.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Prompt Experiments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book-help project tests two **prompt** versions on seven shared development cases.

By the end, I can	How I will show it
Explain the mechanism and terminology of controlled prompt experiments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Prompt Experiments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of controlled **prompt** experiments

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-help project tests two **prompt** versions on seven shared development cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Q improves overall count but worsens missing-source behavior; use the declared priorities.

1. Decide explicitly

2. Paired setup

3. Missing-source slice

4. Overall rates

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Decide explicitly Q improves overall count but worsens missing-source behavior; use the declared priorities.

A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make missing-source correctness a required selection gate. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Neither **prompt** clears a 2/2 gate on these records.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Neither **prompt** clears a 2/2 gate on these records. P fails 7 and Q fails both 4 and 7; both need revision.

Repair a missing building block

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

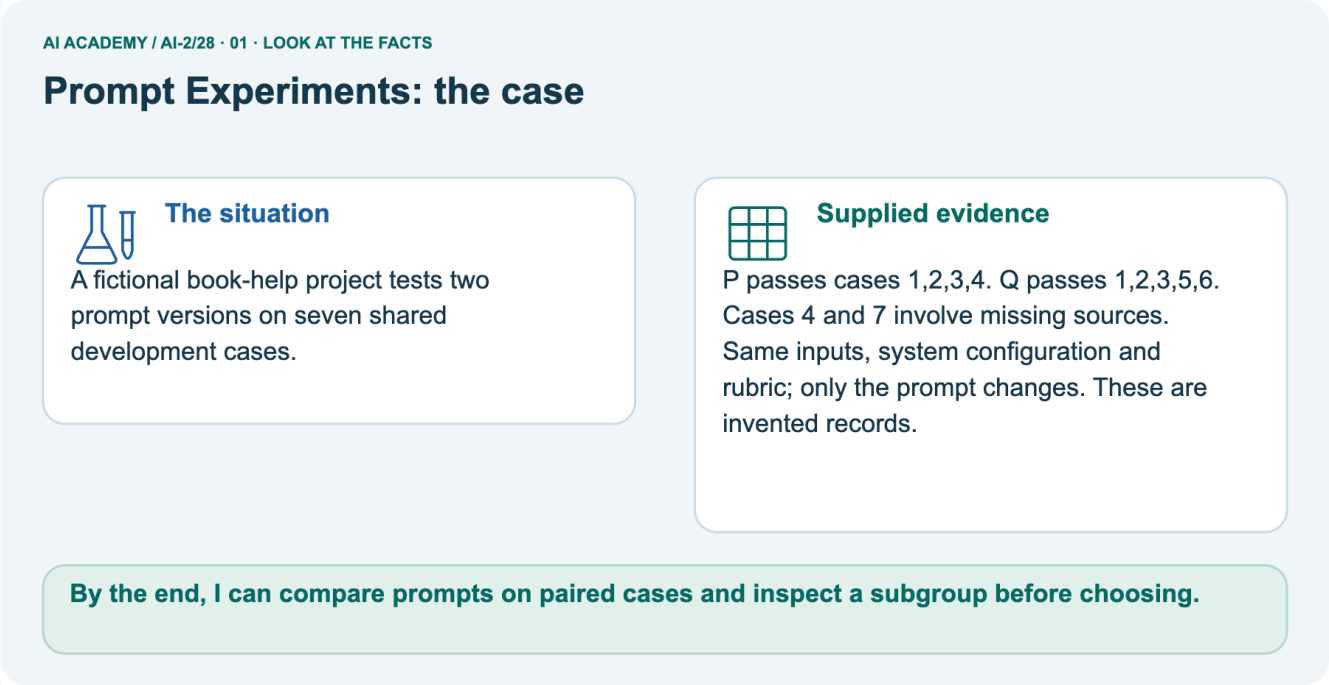
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Prompt Experiments graphic using its labels, deciding evidence and relevant limitation.

01. Prompt Experiments: the case



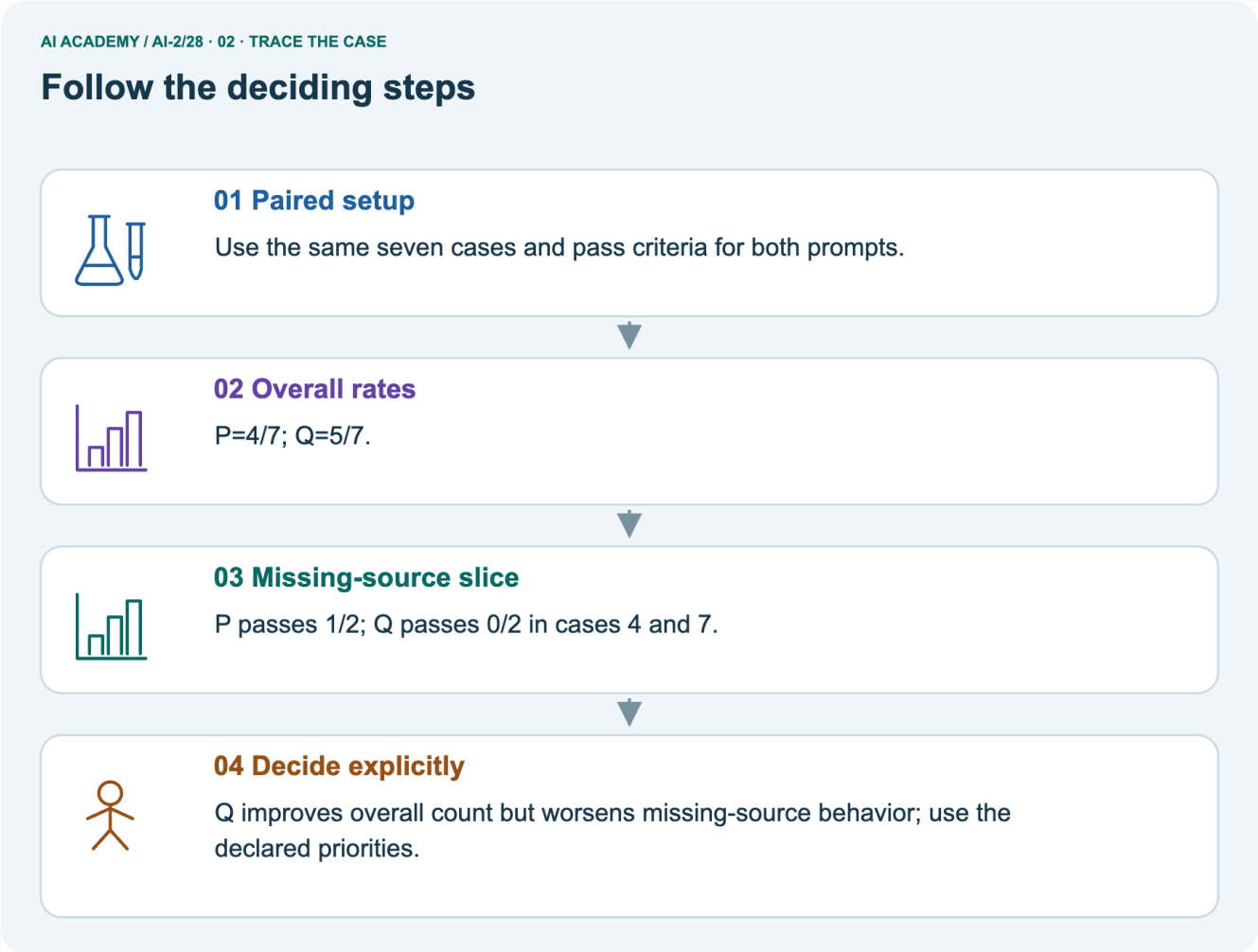
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-help project tests two **prompt** versions on seven shared development cases. Supplied facts: P passes cases 1,2,3,4. Q passes 1,2,3,5,6. Cases 4 and 7 involve missing sources.

Same **input**s, system configuration and **rubric**; only the **prompt** changes. These are invented records.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

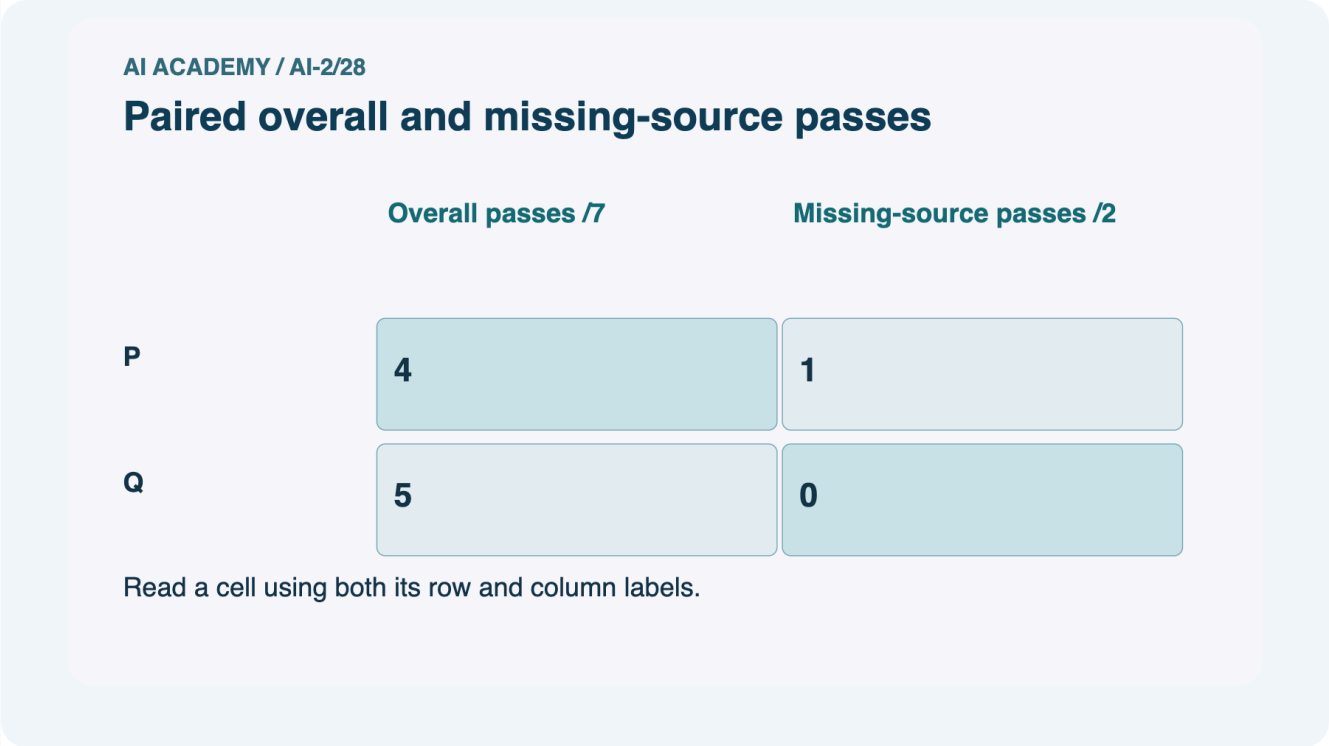


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired setup: Use the same seven cases and pass criteria for both **prompt**s. → Overall rates: P=4/7; Q=5/7. → Missing-source slice: P passes 1/2; Q passes 0/2 in cases 4 and 7. → Decide explicitly: Q improves overall count but worsens missing-source behavior; use the declared priorities.

Point to each numbered stage and explain which recorded **input** it uses.

03. Paired overall and missing-source passes

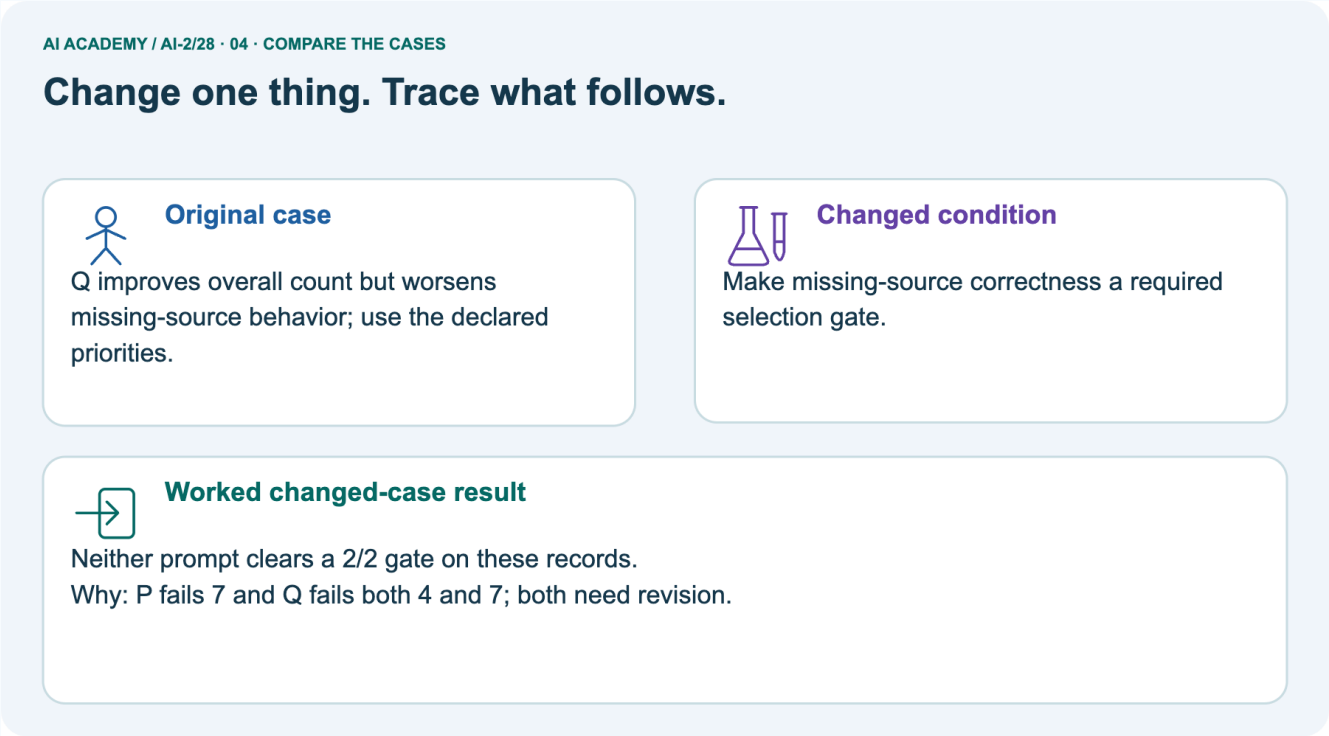


Follow the labelled picture with the matching explanation. Open the vector diagram

Paired overall and missing-source passes. P: Overall passes /7 = 4, Missing-source passes /2 = 1. Q: Overall passes /7 = 5, Missing-source passes /2 = 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



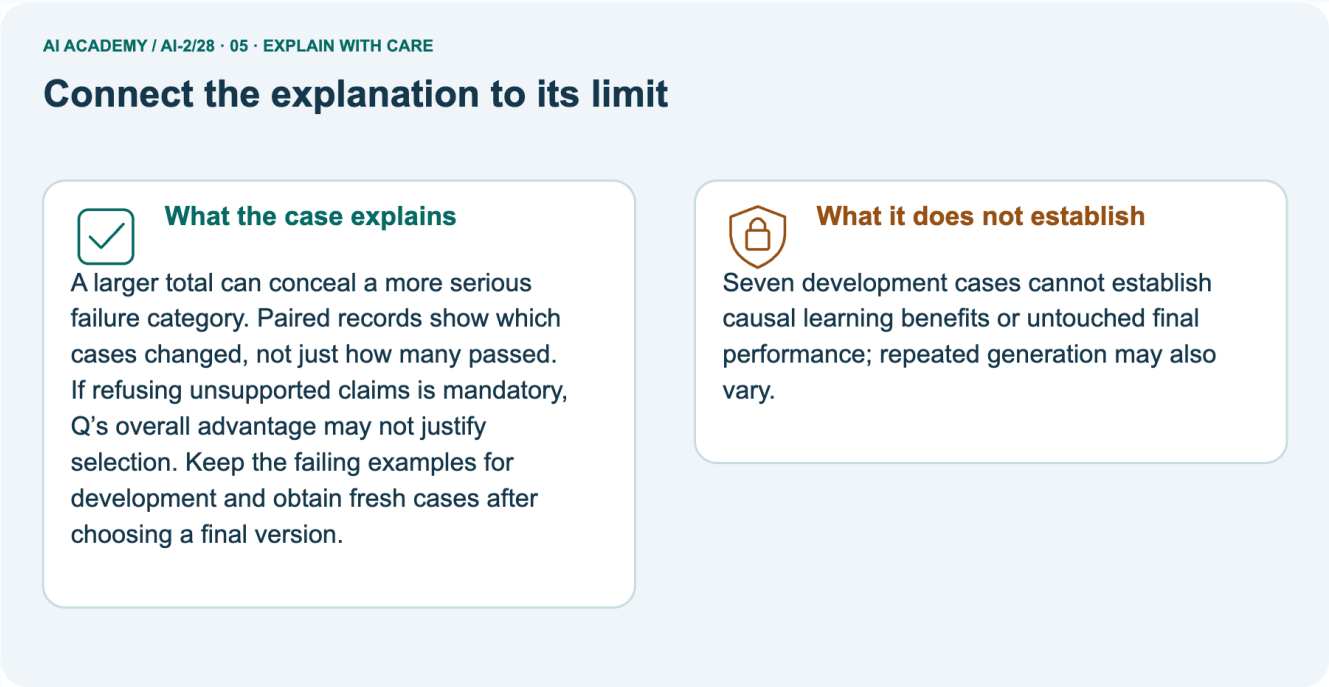
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Q improves overall count but worsens missing-source behavior; use the declared priorities. Changed condition: Make missing-source correctness a required selection gate.

Expected result: Neither **prompt** clears a 2/2 gate on these records. Reason: P fails 7 and Q fails both 4 and 7; both need revision.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

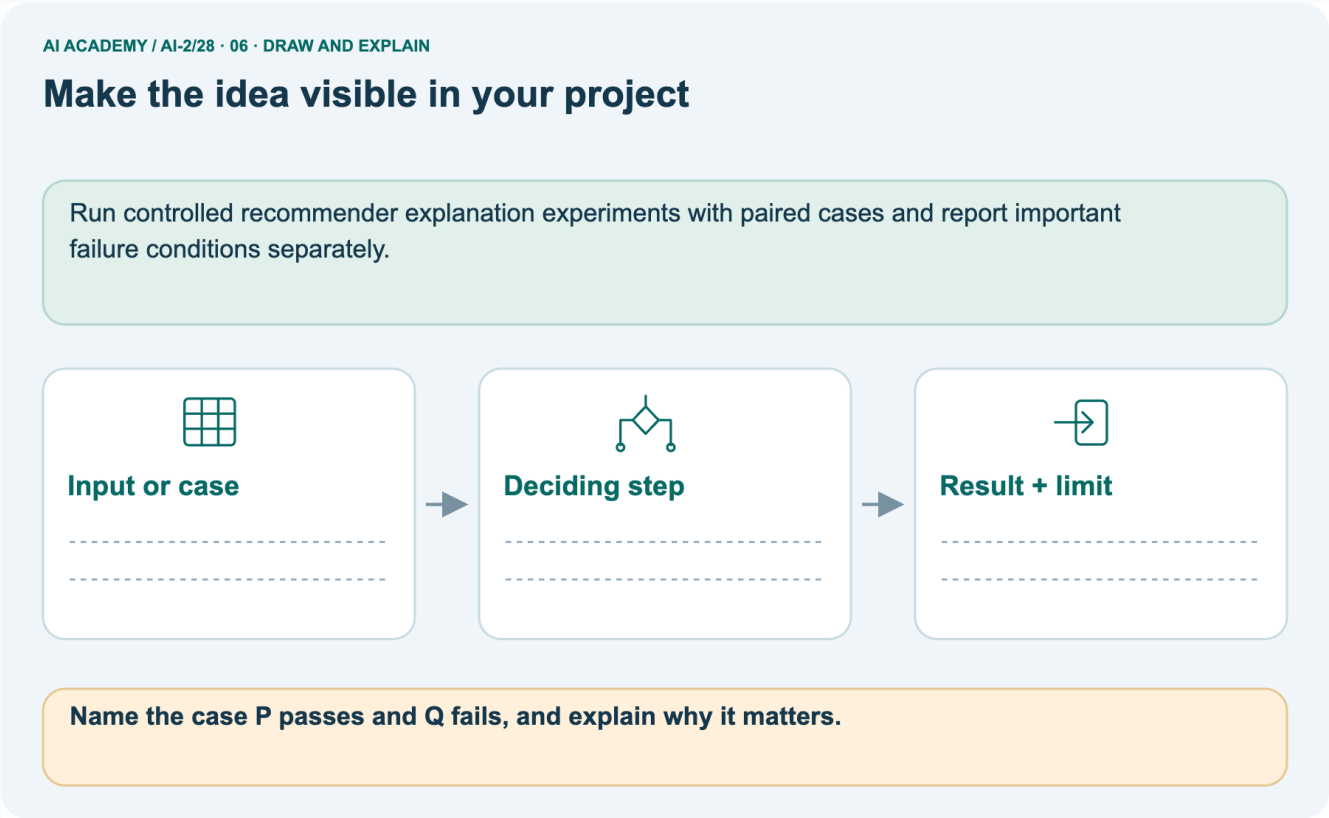
A larger total can conceal a more serious failure category. Paired records show which cases changed, not just how many passed. If refusing unsupported claims is mandatory, Q's overall advantage may not justify selection.

Keep the failing examples for development and obtain fresh cases after choosing a final version.

Limit: Seven development cases cannot establish causal learning benefits or untouched final performance; repeated generation may also vary.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Run controlled recommender explanation experiments with **paired case**s and report important failure conditions separately. Draw your **input**, deciding step, result and limit.

Supported question: Name the case P passes and Q fails, and explain why it matters.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.